Junzhe Shi

443-925-1478 | jshi70@jh.edu | linkedin.com/in/junzheshi

Education

Johns Hopkins University Baltimore, MD
Bachelor of Science in Computer Science, Applied Mathematics and Statistics *Expected May 2028*

- **GPA:** 3.86 / 4.00 Dean's List
- **Mathematics**: Linear Algebra for Data Science, Intermediate Probability and Statistics
- **Computer Science**: Artificial Intelligence, Human/Machine Intelligence Alignment, AI-Enabled Software Engineering, Object-Oriented Software Engineering, Data Structures, Full-Stack JavaScript

Research Experience

Research Assistant, Data Science for Psychiatry Lab February 2026 – Present
Johns Hopkins University *Baltimore, MD*

- Review and synthesize medical and clinical-AI literature to support development of a REACH-based EHR cohort study on thyroid hormone prescribing in psychiatric settings.
- Contribute to research planning for a cohort study comparing thyroid hormone prescribing patterns among antidepressant-treated adults without overt hypothyroidism across psychiatry and primary care.
- Compare Adams-authored thyroid studies with recent work on LLM-assisted diagnostic reasoning, ambient AI, and mental-health AI regulation to evaluate evidence quality, patient safety, risks evaluations, and governance considerations.

Perovskite Solar Cell Research Assistant June 2023 – June 2024
Huazhong Agricultural University *Wuhan, China*

- Collaborated with a 10+ member PVSC research group to optimize next-generation solar cell construction and improve data interpretation.
- Improved 7SCS equipment setup and testing workflows, contributing to an increase in measured solar cell efficiency from approximately 10% to 20%.
- Co-authored a research paper and a review article in *Materials Chemistry C*, contributing to experimental analysis, literature synthesis, and manuscript preparation.
- Presented findings at The 3rd Optoelectronic Materials and Devices Development Forum 2023.

Subatomic Physics Research Leader June 2021 – December 2022
Cathaypath Institution of Science *Guangzhou, China / Online*

- Used ROOT on Ubuntu Linux and the Glauber-Monto-Carlo model to analyze 10,000+ particle collisions, optimizing Pb nucleus collision asymmetry across three harmonic levels and achieving 78.24% accuracy in relating collision number to eccentricity.

Exoplanet Independent Research Initiator June 2022 – December 2023
Cathaypath Institution of Science *Guangzhou, China / Online*

- Used Lightkurve, Matplotlib, and NumPy to analyze TESS luminosity curves and model the planet-star ratio of four major exoplanets in WASP detections with 90.35% accuracy.

Activities & Projects

Atlas – AI-Assisted Course Search & Schedule Planning February 2026 – May 2026
Johns Hopkins University *Baltimore, MD*

- Built an AI-assisted JHU course-search and schedule-planning platform in a full-stack team project.
- Implemented personalized schedule-building workflows across a React frontend and Node.js/Express backend, connecting user preferences, course data, and LLM-assisted advising.
- Integrated PostgreSQL, pgvector, and OpenAI-powered tooling for course discovery and schedule recommendations.

PDR AI / LaunchStack February 2026 – May 2026
Johns Hopkins University *Baltimore, MD*

- Built a multi-step document rewrite workflow in Next.js and TypeScript with file import, preview and diff review, regeneration, and stepwise acceptance of AI edits.

- Expanded the marketing pipeline with trend research across Reddit, X, LinkedIn, and Bluesky, and added rewrite-state persistence plus handoff into the document workspace.
- Integrated DOCX redlining into the legal editor through a validated `/api/legal/apply-edits` route and tests, enabling Track Changes review of AI-assisted edits.

Professional Experience

EchoplusAI Summer Internship - ReferMe May 2025 – August 2025

EchoplusAI *Remote*

- Developed a full-stack web application for users to seek and provide referrals, supporting user registration, login, and profile submission, used by over 200 referral providers and saving the education institute $2,080 in administrative costs.
- Built responsive frontend pages with HTML, CSS, TypeScript, Angular, and Angular Material, including signup, login, referral provider profiles, and profile management workflows.
- Designed and implemented backend services with Node.js and Express.js, including authentication and post services for user login, registration, and profile management.

EchoplusAI Summer Internship - Golang Microservice May 2025 – August 2025

EchoplusAI *Remote*

- Created microservices using Golang, including a broker service for API routing and a logging service for centralized data tracking.
- Deployed the application with Docker and Kubernetes and automated operations using a Makefile, streamlining deployment and service management.

Publications

Trap Engineering Using Oxygen-Doped Graphitic Carbon Nitride for High-Performance Perovskite Solar Cells 2023

- Y. Lei, L. Liu, X. Cai, **J. Shi**, N. Guo, X. Li, Y. Liu, Y. Guo, P. Qin, J. Chen, H. Lei, Z. Tan
- *Materials Chemistry C*, 2023. doi:10.1039/D3TC01711G

Optimization of Asymmetry of Pb-Pb Nucleus Collision Based on Glauber Model Simulation 2023

- **J. Shi**, H. Xu, Z. Song
- *Theoretical and Natural Science*, vol. 28, pp. 206–217, 2023. doi:10.54254/2753-8818/28/20230428

The Relationship Between and Eccentricities Based on Glauber Model 2023

- **J. Shi**
- *Theoretical and Natural Science*, vol. 11, pp. 120–126, 2023. doi:10.54254/2753-8818/11/20230390

Hyperbolic-Tangent-Function-Modeled Transit Light Curve and Planet Radius Calculation 2023

- **J. Shi**
- *Theoretical and Natural Science*, vol. 34, pp. 134–140, 2023. doi:10.54254/2753-8818/34/20240704

Technical Skills

Languages: Java, Python, Go, C/C++, JavaScript, TypeScript, SQL
Frameworks: React, Angular, Node.js, Express.js, Tailwind CSS
Platforms: Windows, macOS, Linux
Developer Tools: Git & GitHub, Docker, Kubernetes, PostgreSQL, MongoDB, SQLite, VS Code, GDB